

PART I. RESERACH

1. **Research** is an *original* investigation undertaken in order to *gain knowledge* and understanding.

- **Originality** is doing or producing something new.
 - *Inductive reasoning* starts with your observations of the world, and comes to general conclusions about it, which is to build models and theories based on your interpretation of the world.
 - *Deductive reasoning* starts with your knowledge and understanding of the world, and predicts likely observations within it, even though you might not have encountered them before.

- **Gain** refers to the fact that research leads to actual contributions of knowledge to yourself and/or others.
- Several concepts:
 - *Data* are the factual elements that describe objects or events. They represent the raw numbers and raw text that you gather from your investigation.
 - *Information* represents data that have been processed in order to provide you with some insight into their meanings.
 - **Knowledge** is your higher level understanding of things. While information provides you with an idea of the 'what', knowledge represents your understanding of the 'why'.

- *Wisdom* represents your ability to put your knowledge into practice. It represents your ability to apply your skills and experiences to create new knowledge and adapt to different situations.
- ★ *Theory* represents ideas, opinions, and suppositions based on your observations of the world. A theory is not necessarily true but, at the moment, it represents the best explanation of what you observe.
- ★ *World knowledge* relates to world understanding, wisdom, and interpretation by everybody and everything that is recorded or documented somewhere and somehow.

2. Research vs. Intelligence-gathering

- Intelligence-gathering refers to the action of collecting data and information on their own.
- Research must go beyond merely gathering data and describing what you see.
 - It must contribute to knowledge, and the type of contribution that research makes based upon its *nature*.
 - *Pure theory*: developing theories to explain things without necessarily linking them to practice.
 - *Descriptive studies*: reviewing and evaluating existing theory and knowledge in a field or describing particular situations or events. This might include testing existing theories, describing the state of the art, or looking for limits in previous generalizations.

- *Exploratory studies*: exploring a situation or a problem. These studies can start out by exploring particularly broad areas, concepts, and ideas before focusing in and narrowing down to specifics as the research progresses.
- *Explanatory studies*: explaining or clarifying something or some phenomena, and identify the relationships between things.
- *Causal studies*: assessing the effects that one or more variables have on another, e.g., 'Does the size of software product affect maintenance difficulty?'.
 - It looks for explanations, relationships, comparisons, predictions, generalizations, and theories.

3. *Characteristics* of good research:

- *Open minds* — to work with an open system of thought. Conventional wisdom and accepted doctrine may turn out to be inadequate.
- *Critical analysis* — examine data critically.
 - Are these figures correct? Have they been affected in some way? What do these data really mean? Are alternative data available? Can these data be interpreted differently?
- *Generalizations* — to generalize and to specify limits on the generalizations.